



Ahasan Ullah Khalid

aukhalid02@gmail.com
+880 1610074044
Dhaka, Bangladesh

Portfolio - aukhalid.vercel.app
GitHub - github.com/aukhalid
LinkedIn - linkedin.com/in/aukhalid

EDUCATION

Chittagong University of Engineering and Technology B.Sc., Electronics and Telecommunication Engineering CGPA: 3.78/4.00 (In Final Year), 4.00 (In all 3 VLSI Courses), 3.10 (Overall)	March 2021 - June 2026 Chittagong, Bangladesh
Dhaka College Higher Secondary Certificate, Science GPA: 5.0/5.0, General Gov. Scholarship	July 2018 – July 2020 Dhaka, Bangladesh
Ideal School and College, Motijheel, Dhaka Secondary School Certificate, Science GPA: 5.0/5.0, Talent Pool Gov. Scholarship	January 2009 – February 2018 Dhaka, Bangladesh

RESEARCH & PUBLICATIONS

B.Sc. Thesis:

- **EVPIX-RV32: 5-Stage Custom RISC-V SoC with Integrated IPU and TinyML Support for Real-Time Edge-Vision AI Acceleration: RTL-to-GDSII Design, Basys-3 FPGA Prototyping, and SkyWater 130-nm CMOS ASIC Implementation.**
Supervisor: Md. Farhad Hossain, Assistant Professor, ETE, CUET

Conference Publications:

- A. U. Khalid, A. Tabassum, A. I. Rupom, M. F. Hossain and M. A. Hossain, "**Design and Performance Comparison of 4-bit Ripple-Carry and Carry-Look-Ahead Adder in 90-nm CMOS VLSI Technology,**" Published in *Proc. 2nd IEEE QPAIN, April 2026*
IEEE Xplore Digital Library link

WORK EXPERIENCE

ADN Semiconductors Ltd. RTL Design and Verification Engineer (Intern) • RTL Design Projects with SystemVerilog HDL, verifying them with testbench and full ASIC/FPGA implementation for different clients.	May 2026 – Present Intern
Manarat International University (MIU) VLSI Lecturer • Delivering comprehensive lectures and practical sessions on VLSI design for the EEE department, developing quizzes and examination questions, evaluating student assignments and assessments.	May 2026 – Present Part-Time

TECHNICAL SKILLS

- **Softwares:** Cadence Virtuoso, OpenROAD, Vivado, Quartus, Logisim Evolution, MATLAB
- **Languages:** Verilog HDL, SystemVerilog, C, C++, Python, SQL, HTML, CSS, JavaScript, TypeScript, React, Tailwind, Node JS, Express JS
- **Frameworks:** React, Tailwind, Node JS, Express JS
- **Management:** Google Apps, Microsoft (Word, Excel, PowerPoint etc.), Notion
- **OS:** Linux, Windows

PROJECTS

VLSI Project: CMOS VLSI Logic Designs gpdk90

- CMOS standard cells and digital system designs were implemented using the GPDK90 open-source 90-nm PDK in Cadence Virtuoso. Includes transistor-level schematics, layouts and simulations.

VLSI, RTL & FPGA Project: RISC-V RV32I CPU Core Design and Verification

- Complete RTL design and verification of a RISC-V RV32I CPU Core. The whole design was done on Vivado using SystemVerilog Language. The verification and testing is also done on SystemVerilog. Basys 3 FPGA was used to implement the project on hardware level.

VLSI & DLD Project: SAP-1 CPU in Logisim

- Complete implementation of the Simple-As-Possible (SAP-1) CPU architecture designed and simulated using Logisim Evolution. This project features a hardwired Control Unit, ROM and a Bootloader to automatically load data into the RAM.

IoT Project: AI Agent Powered Smart Home Automation and Fault Aware Load Control

- Implemented AI-powered smart home automation system using ESP32, MQTT, and multiple sensors for real-time monitoring and safety management. It features autonomous emergency responses and a Groq Llama-3.3 based AI agent for natural language home control.

Web Design Project: Nokshia

- Implemented a front end web design project of a company called "Nokshia". Used HTML, CSS, Tailwind etc for this project. The main purpose of this project is the design aesthetics of the company.

TRAINING & COURSES

Unive VLSI Mentorship Training Program

October 2025 – February 2026

Trainee

- Completed full VLSI Training program and learned electronics, digital logic design, VLSI analog and digital IC design etc and learned how the industry works.

Teletalk Bangladesh Ltd.

April 2025 – May 2025

Intern

- Completed industrial attachment and learned about how the networking works in circuit switching and packet switching.

EXTRACURRICULAR & LEADERSHIP

CUET Chess Club

July 2025 – Present

President

- Organized tournaments, led club team and hosted regular sessions.

AshesToFlame Records

February 2019 – 2023

Founder / Co-Owner

- Managed music production, release workflow, branding, collaborations, artists/talents management.

ETE Televerse 1.0

November 2025

Branding & Promotion Head

- Managed branding and promotion for the mega event ETE Televerse 1.0 and got 200,000 interactions and impressions on social media, also produced the theme song "Eksathe" for this event.

CUET Debating Society

April 2022 – May 2025

Debater

- Runners up of IDDC (Inter Department Debate Competition) in 2022 at CUET

INTERESTS

Music Production, Gaming, Traveling | Reading, Watching Movies, Series | Anime, Esports, Sitcom, Chess

REFERENCES

Md. Farhad Hossain

Assistant Professor, Department of ETE
Chittagong University of Engineering and Technology
(CUET)

Email: farhad.hossain@cueta.ac.bd

Phone: +880 1601-998766

Arif Istiaque

Assistant Professor, Department of ETE
Chittagong University of Engineering and Technology
(CUET)

Email: arif.ete@cueta.ac.bd

Phone: +880 1521-331635